Samsung

Samsung operates in a market environments characterized by high capital intensity, rapid innovation, and structural interdependence. In the global smartphone sector Samsung is in the premium market segment. Each flagship product release cycle forces executive leadership into a strategic dilemma maintaining high premium prices maximizes gross profit margins per unit where lowering prices captures market share from the competitor but risks initiating a value destroying price war.

While traditional management frameworks such as SWOT or Porter's Five Forces provide a static overview of industry competitiveness and Game Theory supplies a rigorous mathematical framework to model competitor interdependence, evaluate best responses, drive equilibrium and anticipate rival reactions in dynamic pricing environments.

In today's highly competitive tech industry two firms Samsung and Apple consistently dominate global smartphone sales. Both companies regularly face a strategic dilemma how should they price their products in a market where the rival's actions significantly impact their outcomes. If Samsung lowers its prices it could gain market share at the expense of profits. If both firms lower prices, a price war ensues. This strategic interaction makes pricing decisions an ideal context for applying game theory to a branch of mathematics that do the decision making.

This internal assessment uses game theory to model pricing decisions between Samsung and Apple. Specifically It will construct a payoff matrix representing the potential outcomes of two competing pricing strategies High Price and Low Price.

- Core Mathematical Building Blocks of the Game

Firm: The rational independent decision makers in the market are Samsung and Apple. Both players act strategically to maximize their own earnings.

Strategy Sets: The set of possible pricing choices available to each player is High Price (HP) and Low Price (LP). High Price Charge premium prices to maintain high profit margins per phone sold. Low Price (LP) or offer discounts to sell higher volumes and gain market share.

Payoffs: The net quarterly operating profits in millions of dollars earned by each firm determined by the combination of choices made by both companies.

Dominant Strategy: A strategy that gives a player a strictly higher profit than any other option regardless of what choice the competitor makes.

Nash Equilibrium: A outcome of Samsung and Apple where neither of this company has an incentive to change its decision on its own. At this point any single firm changing their choice unilaterally would result in a lower profit.

This theoretical foundation provides the tools necessary to mathematically model the strategic decisions made by Apple and Samsung.

- Theoretical Payoff Structure

To simulate a simplified pricing scenario I consider Samsung and Apple competing each with two strategic choices:

High Price (HP): Maintain premium pricing

Low Price (LP): Offer price cuts to capture market share

Their profits depend not only on their own decision but also on what the other company chooses. For this model. I assume both companies release competing flagship phones in the same quarter for e.g, iPhone vs Galaxy S series.

The payoffs are hypothetical but reflect realistic business behavior and price sensitivity:

Apple \ Samsung	Samsung: High Price (HP)	Samsung: Low Price (LP)
Apple: High Price (HP)	(8, 8)	(3, 10)
Apple: Low Price (LP)	(10, 3)	(5, 5)

This table uses game theory to model how Samsung and Apple decide their product pricing strategies. The numbers in each box represent profit scores where the first number belongs to Apple and the second to Samsung. If both companies keep their prices high, they split the market fairly and both earn strong profits of 8. However, if one company keeps its price high while the other cut prices the cheap seller most of the market and earns a maximum profit of 10 leaving the high priced seller with a low payoff of 3.

Because of this setup, both companies face a powerful temptation to undercut each other. If Samsung sets high prices Apple makes more money by setting low prices earning 10 instead of 8. If Samsung sets low prices Apple is still better off setting low price earning 5 instead of 3 to avoid being crushed. Since the Samsung goes through the exacts same logic charging to the low price becomes the dominant individual strategy for both companies no matter what the other rival does and do.

This creates a classic economic trap known as the Prisoner's Dilemma. Because neither company can trust in each other and not to discount, both necessarily choose low prices and set it into an outcome where each earns a modest profit of 5. Even though both firms would be significantly better when they cooperating with each other and get at a high prices to make 8 each individual self interest and fear of being undercut force them into a price war that losses profits from the both sides.

- Buyer-Supplier Game

Dual Relationship Structure:
Downstream Competition: Direct rivalry in high end flagship smartphones like iPhone series vs Galaxy series.
Upstream Cooperation: Apple relies on Samsung Display and Semiconductor divisions for critical OLED displays, DRAM, and NAND flash memory.

Entity	Target / Recipient	Relationship Type	Action / Interaction
Apple Inc.	Samsung Electronics	Supply Chain, B2B	Purchases and Revenue
Samsung Electronics	Apple Inc.	Component Supplier	Component Supply
Apple Inc.	Samsung Electronics	Legal, Intellectual Property	Lawsuits

Strategic Incentive Conflict:
Apple Incentive: Threaten legal action or diversify supply sources like for e.g, TSMC, LG Display to avoid over reliance on a direct consumer competitor.

Samsung Incentive: Maintain component sales to Apple generating billions in component revenues while defending its Galaxy smartphone market share.

Outcome: Samsung component division acts as an internal hedge against losses incurred by its mobile handset division during legal battles or market share swings.

A strategy combination forms a Nash Equilibrium if both players simultaneously play best responses to each other. Since the Samsung always chooses low price, Apple best responds is by also choosing low price.

- Algebraic Derivation of Mixed Strategy Equilibrium

To explore whether randomized pricing strategies exist in equilibrium we drive the Mixed Strategy Nash Equilibrium. In a mixed strategy players assign probabilities to their pure strategies such that the opponent becomes indifferent between their choices.
Variable Definitions:
Let p = Probability that Apple plays High Price, so (1 - p) = Probability Apple plays Low Price.
Let q = Probability that Samsung plays High Price, so (1 - q) = Probability Samsung plays Low Price.

To make Apple indifferent between playing High Price and Low Price its expected payoffs from both strategies must be equal. In game theory a player is willing to choose between different choices only when every available option yields the exact same average return. This calculation finds the precise probability with which Samsung must choose its pricing strategy to make Apple neutral between charging a high price or a low price.

Expected Payoffs: Apple expected return from setting a high price is a weighted average of its possible gains depending on how likely Samsung is to set a high price vs a low price. Similarly Apple expected return from setting a low price is weighted by those exact same chances.

The Indifference Condition: To make Apple indifferent the overall expected return from choosing a high price is set equal to the overall expected return from choosing a low price.

Solving for Samsung's Strategy: Setting these two expected-payoff expressions equal to each other creates an algebraic relationship where the only unknown element is Samsung probability of choosing a high price. Simplifying the equation isolates that single probability.

Strategic Outcome: When Samsung prices high at this exact rate Apple earns the same overall benefit whether it prices high or low. Because neither option offers a better result than the other Apple has no incentive to favor one price point over the other fulfilling the condition for a mixed-strategy equilibrium.

- Comprehensive Evaluation of Model Strengths vs. Limitations

Model Strengths	Model Limitations
Explicitly incorporates competitor interdependence into decision-making.	Binary strategy reduction (real pricing uses continuous elasticity curves).
Sensitivity testing demonstrates structural stability of equilibrium.	Static, simultaneous assumption (ignores multi-period release timing).
Mathematically proves why rational firms fall into price war traps.	Remove non-price factors (iOS ecosystem lock-in, brand loyalty).
Grounded in empirical ASP, unit cost, and market share data.	Assumes symmetric information regarding cost structures and margins.

Conclusion: To a high extent game theory effectively models the competitive dynamics and pricing traps between Apple and Samsung. It proves that non-cooperative pricing pushes both firms into a suboptimal Prisoner's Dilemma.

Reference

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